

Python Hands-On Questions (Strings, Lists, Tuples, Sets, Dictionaries)

STRING QUESTIONS

1. String Method Drill: Perform upper(), replace spaces, count vowels, slice first 3 & last 3 chars.
2. Substring Extraction: Extract alternate characters from a string using slicing.
3. Palindrome-like Variation: Check reversed string match ignoring case & non-alphabets.
4. Character Frequency Finder: Build dict of char counts manually.

LIST QUESTIONS

5. List Slicing Gym: Even indices, last 5 elements, reverse using slicing.
6. List Methods Lab: Use append, insert, extend, pop, remove with 5 inputs.
7. Sort by Last Name: Sort list of names by last name using sort().
8. List Comprehension: Generate squares, evens, >avg numbers.
9. 2D List Slicing: Extract column, diagonal, middle row.
10. Unique Pairs: Create unique (a,b) pairs using list comprehension.

TUPLE QUESTIONS

11. Tuple Unpacking: Unpack tuple into variables & print formatted.
12. zip() Challenge: Zip names & scores -> list, dict, highest scorer tuple.
13. Tuple Immutability: Try modifying tuple, handle error, create modified new tuple.

SET QUESTIONS

- 14. Remove Duplicates with Sets: Maintain order while deduplicating.
- 15. Set Operations: Find only live, only recording, both, none.
- 16. Membership Checker: Check banned words in sentence.
- 17. Symmetric Difference: Errors appearing in only one log.

DICT QUESTIONS

- 18. Dictionary CRUD Lab: Create dict from input, get, update, delete, print keys/values/items.
- 19. Nested Dictionary Access: Access dept, names, oldest age.
- 20. Dict Comprehension: {num:num*num}, {num:'even/odd'}, {char:ASCII}.